

DATA ANALYTICS PROJECT

Churn Analysis and Customer Intelligence



By Shubham Verma

Churn Analysis & Customer Intelligence

1

The Business Challenge

In the hyper-competitive OTT landscape (Netflix, Hotstar, Prime), retention is the only way to survive. You'll act as a Data Analyst identifying high-risk subscribers using a multi-dimensional dataset — customer demographics, subscription tiers, and support escalations.

2

Core Tech Stack

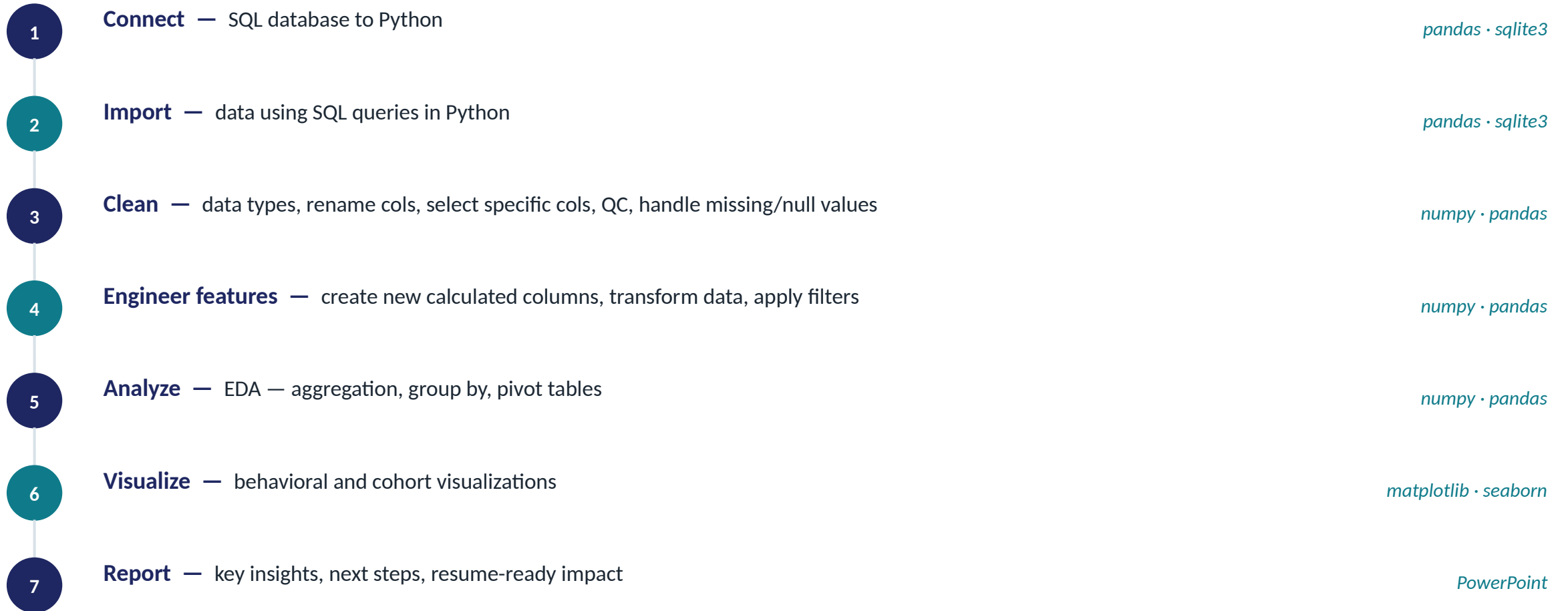
SQL & Python integration (numpy, pandas, sqlite3, matplotlib, seaborn) · Performance engineering (data cleaning, feature engineering, analytics) · Behavioral visualization · Writing actionable insights.

3

Project Milestones

Relational Data Extraction: connecting Python to SQL databases to pull multi-table datasets.
Advanced Feature Engineering: data imports, calculating tenure, churn rates, and customer aging.
Executive Reporting: translating technical findings into high-impact business insights.

Churn Analysis — Roadmap



Churn Definition by Business

Churn analysis is the process of figuring out why customers stop doing business with you.

Who

Identifying which customers have left, or are likely to leave.

Why

Analyzing their behavior in the lead-up to departure.

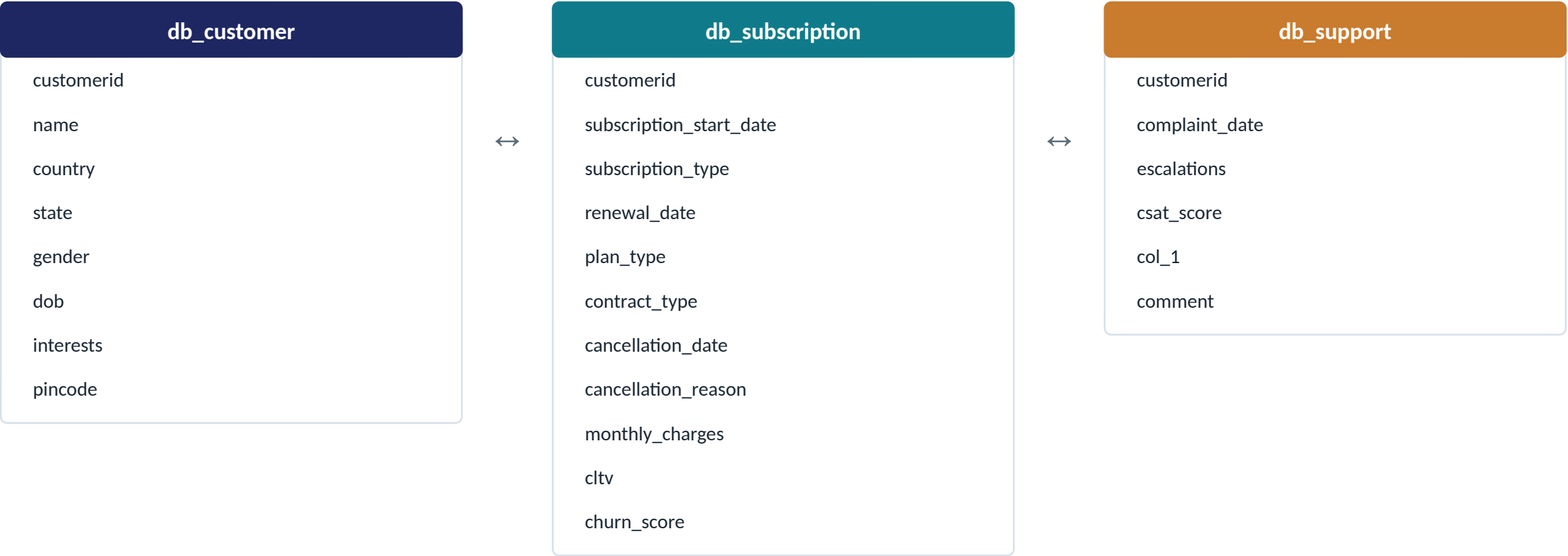
When

Finding the “danger zone” — the point at which a user is likely to leave.

Business Type	Churn Definition
SaaS	Subscription canceled
E-commerce	No purchase in 90 days
Adtech	Not using service/app for 90–120 days
Streaming	Membership inactive
Telecom	Account terminated
Banking	No transactions for X months

Database Tables

Database: customer_churn



All three tables join on customerid, giving a single 360° view of each subscriber across demographics, subscription behavior, and support history.

Calculated Metrics

KPI	Formula
Churn rate	<i>Churned customers / Total customers</i>
Churn by plan type	<i>Churn rate GROUP BY plan_type (Basic / Standard / Premium)</i>
Churn by state	<i>Churn rate GROUP BY country, state</i>
Retention rate	<i>1 – Churn rate</i>
ARPU	<i>SUM(monthly_charges) / COUNT(active customerid)</i>
Average customer tenure	<i>AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))</i>
Revenue at risk	<i>SUM(monthly_charges) WHERE churn_score > 70</i>
Escalation rate	<i>SUM(escalations) / COUNT(complaints) × 100</i>
Avg complaints per customer	<i>COUNT(complaints) / COUNT(DISTINCT customerid)</i>
Correlation: escalations → churn	<i>Churn rate WHERE escalations ≥ 1 vs. 0</i>

Insights & Action Items

Insights

28.6%

Churn rate

71.4%

Retention rate

1,451 days

Avg. tenure

₹18.80

ARPU

₹395

Total revenue

₹74 (18%)

Revenue loss (churn)

₹2,047

CLTV lost

55.6% vs. 8.3%

Monthly vs. annual churn

Most churn comes from the Basic plan — limited revenue impact so far, but worth monitoring.
Most churn occurred in September 2024, concentrated in Karnataka.

Action Items

- Investigate Karnataka: check for recent price increases, complaints, or technical issues.
- Confirm whether the Basic plan's subscription price changed, especially around September.
- Review competitor activity — at least one customer switched to a competitor.
- Prioritize High and Medium risk customers by LTV, review complaint history, and reach out via email, SMS, and calls to resolve their issues.

Portfolio Project — Summary

Churn & Revenue Impact *(General Summary)*

Engineered an end-to-end churn analytics pipeline for an OTT subscription platform by integrating multi-table subscriber data across acquisition type, contract structure, and plan tier (20+ KPIs). Uncovered a significant churn disparity between monthly and annual contract segments, quantified MRR leakage and CLTV erosion attributed to high-risk cohorts, and delivered a data-backed contract-migration retention strategy.

Risk Scoring & Segmentation *(Risk Analysis & Segmentation)*

Developed a multi-dimensional churn risk scoring model by synthesizing subscription tenure, plan type, and support escalation signals across three relational tables (20+ KPIs), segmented the customer base into risk tiers using composite churn scores, exposed a significant lifetime value gap between churned and retained cohorts, and recommended prioritizing Premium annual-plan retention to maximize long-term revenue yield.

Support Intelligence & Cancellation Analysis *(Working on Live Data)*

Performed cross-functional support-churn correlation analysis by joining complaint, escalation, and CSAT data with subscription records, identified that escalated support interactions were disproportionately concentrated among churned customers, decomposed cancellation drivers into competitor switching, pricing sensitivity, and content dissatisfaction, and translated findings into a prioritized product and pricing roadmap for the platform's growth team.

Churn & Revenue Impact

Engineered an end-to-end churn analytics pipeline for an OTT subscription dataset (20+ KPIs), identifying a **28.6% overall churn rate** and surfacing that monthly-contract subscribers churned at **55.6%** — 6.7× the **8.3%** annual-contract rate — directly attributing **₹73.94/mo** in MRR leakage and **₹2,047** in CLTV erosion to six at-risk customers, enabling a targeted contract-migration retention strategy.

28.6%

Churn Rate

55.6% vs 8.3%

Monthly vs Annual Churn

₹73.94/mo

MRR Leakage

₹2,047

CLTV Erosion

Tech Stack at a Glance



SQLite

Relational data storage & querying



pandas

Data import, cleaning & feature engineering



NumPy

Numerical computation & array operations



Matplotlib

Foundational plotting & charts



Seaborn

Statistical & behavioral visualization



PowerPoint

Executive-ready insight reporting